Assuming breach reverses the assumption: everyone is untrusted until the system says they are.

Authorizing every access doesn’t mean just the initial access to a particular system, but every request, even in the same session. This puts an incredible load on the authorization system, requiring speed and flexibility. PBAC systems offer performant fine-grained access control based on the context of each request, making zero trust possible.

1.3.3 The Internet of Things

More and more devices are smart, connecting to the cloud to provide functionality that their dumb counterparts can’t. The rise of the Internet of Things (IoT) means that more aspects of our daily lives—and even entire industries—are now managed by interconnected systems that can communicate and act on our behalf. I have a smart lock, for example. The access control system that controls it literally controls physical access to my house.

However, these devices can also become entry points for cyberattacks. If compromised, smart devices can serve as conduits for hackers to infiltrate more critical systems. Restricting their access to other systems based on their real access needs protects them and prevents them from becoming a vector that hackers can exploit to break into other sensitive systems.

Since many smart devices operate at the edge, they may operate autonomously, working without direct oversight. Dynamic authorization can allow devices to respond to events in a secure, flexible manner, even without a network connection.

This need is especially critical in large, heterogeneous smart device deployments, like in factories or large buildings, where diverse IoT systems must coordinate securely. Here, access control requires real-time decision-making based on device attributes, policies, fluctuating risk levels, and other environmental factors. Traditional static models like ACLs or RBAC struggle to provide the necessary flexibility, making policy-based access control a more effective approach to securing modern IoT environments.